

Chapter 9

1. What index value does the third element of an array have?

The index value of the third element in an array is 2. This is because indexes start at 0.

2. Write the declaration for an array named quantities that store 20 integers.

```
int[] quantities = new int[20];
```

3. Write a declaration for an array named heights storing the numbers 1.65, 2.15, and 4.95

```
double[] heights = {1.65, 2.15, 4.95};
```

4. Write a for-each statement that displays the integer values stored in an array named grades

```
for (int grade : grades)
{
    System.out.println(grade);
}
```

6. How does passing an entire array to a method differ from passing a single element of the array?

Passing an entire array to a method gives the method access to all elements in the array. Passing a single element of the array passes only one value from the array.

7. Why are offset array indexes required in some cases?

Offset array indexes are required in some cases when you want to start reading or writing from any position except 0.

8. What output is displayed by the statements below?

```
String name = "Elaine";
System.out.println(name.charAt(3));
```

The output will display "i". This is because index 3 will show the 4th letter in Elaine, as indexes start at 0. Therefore at charAt(3), it will show the letter "i"

10. Give an example of when a dynamic array might be a better structure choice over an array.

A dynamic array can be better when the number can change during the program.

11. How does the ArrayList indexOf() method determine equality between the object passed to the method and an element in the array?

The indexOf() method determines equality by using the equals() to determine equality between the object passed to the method and an element of the array.

12. How can the values of wrapper class objects be compared?

The values of the wrapper class can be compared by using the compareTo() method.